

Network Traffic Analysis using Machine Learning

A Project Report

**Submitted in the Partial Fulfillment of the Requirements for
The Award of the degree**

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)**

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CERTIFICATE

This is to certify that this project work entitled “**Network Traffic Analysis using Machine Learning**” is a Bonafide record of project work done **A. USHA SRI (228B1A4401), V. VENKATA SOWMYA (228B1A4431), SK. ABIDA ASHRAF (228B1A4426), SK. KOWSAR (228B1A4427)**, in partial fulfillment of the requirements for **IV Year - II Sem** of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)** from **JNTUK**, during the period of **2025 - 26**, is a record of Bonafide work carried out by him/her under our esteemed guidance and supervision.

PROJECT GUIDE

HEAD OF THE DEPARTMENT

EXTERNAL EXAMINER

DECLARATION

We here, by declare that the work is being presented in this dissertation entitled “**Network Traffic Analysis using Machine Learning**” submitted towards the partial fulfilment of requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)** work carried out under the supervision of **Mr. P. Isaac Paul** Placement Training Officer, **Rise Krishna Sai Gandhi Group of Institutions, valluru**. The results embody in this dissertation report have not been submitted by me for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

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ABSTRACT

With the rapid growth of internet-based services, cloud computing, and digital communication, modern network infrastructures are experiencing an unprecedented increase in data traffic volume, speed, and complexity. This rapid expansion has also led to a significant rise in cyber threats such as intrusion attempts, malware propagation, and denial-of-service attacks, making network security a critical concern for organizations and service providers. Traditional rule-based and signature-based network monitoring systems are limited in scalability and adaptability, as they rely on predefined rules and known attack signatures, which makes them ineffective against evolving and unknown threats. This paper presents a machine learning-based approach for network traffic analysis to automatically classify traffic as normal or abnormal. The proposed system utilizes flow-based features extracted from network traffic datasets and applies machine learning algorithms such as Decision Tree, Random Forest, and Support Vector Machine for effective traffic classification. By learning patterns from historical traffic data, the system can detect anomalies and potential security threats with higher accuracy and reduced manual intervention. Experimental results demonstrate that machine learning-based traffic analysis significantly improves detection performance, scalability, and adaptability, making it suitable for deployment in real-world network security environments.

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1. INTRODUCTION

In today's interconnected digital landscape, the volume, diversity, and complexity of network traffic have escalated exponentially, presenting unprecedented challenges for network administrators and cybersecurity professionals. The traditional rule-based approaches to network traffic analysis have proven inadequate in effectively identifying and mitigating emerging threats and anomalies.

Consequently, there is a growing imperative to leverage advanced techniques such as Machine Learning (ML) to augment the capabilities of network traffic analysis systems. Machine Learning offers a paradigm shift in network traffic analysis by enabling systems to learn patterns, detect anomalies, and make predictions from data without explicit programming. ML algorithms can analyze vast amounts of network data, identify subtle patterns indicative of malicious activities or performance issues, and adapt to evolving threats in real-time. The integration of ML in network traffic analysis holds immense promise across various domains, including network security, performance optimization, and resource allocation. ML algorithms can facilitate traffic classification, intrusion detection, anomaly detection, and predictive analysis, thereby enhancing the resilience and efficiency of network infrastructures.

However, while ML presents significant opportunities, its adoption in network traffic analysis poses several challenges. These include the need for labeled datasets for training supervised learning models, the complexity of feature engineering in network data, and the interpretability of ML models for effective decision-making. In this paper, we delve into the application of Machine Learning techniques in network traffic analysis. We explore the diverse range of ML algorithms and methodologies employed for different tasks, ranging from traffic classification to predictive analysis. Additionally, we discuss the preprocessing techniques essential for data preparation and feature extraction, as well as the significance of labeled datasets in model training.

1.1 MOTIVATION

The rapid growth of internet usage, cloud computing, and connected devices has significantly increased the volume and complexity of network traffic. Traditional rule-based and signature-based systems struggle to analyze large-scale network data and detect new or evolving cyber threats effectively. These systems often generate high false positives and require continuous manual updates. Machine Learning provides an intelligent approach by automatically learning patterns from network data and identifying anomalies or malicious activities in real time. Therefore, this project is motivated by the need to develop an automated and efficient system that uses Machine Learning techniques to analyze network traffic and improve network monitoring and security.

1.2 PROBLEM DEFINITION

Modern computer networks generate huge amounts of traffic data through various protocols such as TCP, UDP, and DNS, making manual analysis extremely difficult. Traditional traffic analysis methods are limited in their ability to detect unknown attacks, adapt to new traffic patterns, and handle large datasets efficiently. As a result, network administrators face challenges in accurately identifying abnormal traffic behavior and potential security threats. The main problem addressed in this project is the need for an intelligent system that can automatically analyze network traffic data, classify protocol types, and detect abnormal patterns using Machine Learning algorithms.

1.3 OBJECTIVE OF PROJECT

The main objective of this project is to develop a Machine Learning-based network traffic analysis system that can automatically analyze and classify network traffic data. The project aims to preprocess network datasets, train machine learning models such as K-Nearest Neighbors (KNN) and Naïve Bayes, and evaluate their performance using metrics like accuracy, precision, recall, and F1-score. By comparing the performance of these algorithms, the system identifies the most effective model for predicting network protocol types and analyzing network behavior. Ultimately, the goal is to create a reliable and efficient solution for improving network traffic monitoring and security.

2. LITERATURE SURVEY

The integration of Machine Learning (ML) techniques into network traffic analysis has garnered significant attention in recent years due to its potential to enhance the effectiveness and efficiency of network security and performance management systems. A review of the existing literature reveals a plethora of research studies, methodologies, and applications exploring various aspects of ML in network traffic analysis.

One of the primary areas of focus in the literature is the application of ML algorithms for traffic classification. Numerous studies have investigated the effectiveness of supervised learning algorithms such as Support Vector Machines (SVM), Random Forests, and Neural Networks in accurately categorizing network traffic into different classes, such as web browsing, email, file transfer, and streaming media. These studies often emphasize the importance of feature selection and data preprocessing techniques to improve classification accuracy and reduce computational overhead.

In the realm of intrusion detection, ML-based approaches have emerged as a promising alternative to signature-based detection systems. Researchers have explored the use of anomaly detection algorithms, including clustering techniques, autoencoders, and ensemble methods, to identify deviations from normal network behavior indicative of malicious activities. Additionally, hybrid approaches combining signature-based and anomaly-based detection methods have been proposed to enhance the detection capabilities of intrusion detection systems.

An essential aspect of ML-based network traffic analysis is the availability of labeled datasets for model training and evaluation. Several studies have focused on the creation and curation of labeled datasets, either through manual annotation or automated labeling techniques. Additionally, efforts have been made to address the challenge of class imbalance in network traffic datasets, where certain classes may be underrepresented, leading to biased model performance.

Furthermore, research in the literature has highlighted the importance of interpretability and explainability in ML-based network traffic analysis. As ML models become increasingly complex,

understanding the rationale behind model predictions and decisions is crucial for network administrators and cybersecurity analysts. Techniques such as feature importance analysis, model

visualization, and post-hoc explanation methods have been proposed to enhance the interpretability of ML models in network traffic analysis.

Despite the significant progress made in ML-based network traffic analysis, several challenges and limitations persist. These include the need for real-time processing of large-scale network data, the vulnerability of ML models to adversarial attacks, and the ethical implications of data privacy and security. Addressing these challenges requires interdisciplinary collaboration between researchers in machine learning, network security, and data privacy.

In conclusion, the literature review underscores the growing interest and research efforts in leveraging Machine Learning for network traffic analysis. By addressing key challenges and exploring innovative methodologies, ML has the potential to revolutionize the way network traffic is monitored, analyzed, and secured in an increasingly interconnected digital world.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the existing system of network traffic analysis, traditional methods predominantly rely on rule-based approaches and signature-based detection systems to identify known threats and anomalies within network traffic. These systems often struggle to keep pace with the dynamic and evolving nature of cyber threats, as they rely on predefined rules or signatures that may not encompass the full spectrum of potential attacks. Additionally, manual analysis and human intervention are typically required to investigate suspicious activities, leading to delays in threat mitigation and response. Moreover, the scale and complexity of modern network infrastructures pose significant challenges for traditional approaches, as they may struggle to handle the vast volumes of data generated by diverse sources such as IoT devices, cloud services, and mobile applications. As a result, there is a pressing need for more efficient and adaptive solutions capable of autonomously analyzing network traffic in real-time and detecting emerging threats without human intervention.

DRAW BACKS:

In the existing system of network traffic analysis, several drawbacks and limitations persist, predominantly associated with traditional rule-based approaches and signature-based detection systems:

- 1. Limited Adaptability:** Rule-based and signature-based systems are inherently limited by their reliance on predefined rules or patterns. As a result, they struggle to adapt to new and evolving threats that may not conform to existing rules or signatures. This lack of adaptability makes these systems vulnerable to zero-day attacks and sophisticated evasion techniques employed by malicious actors.
- 2. High False Positive Rates:** Traditional systems often suffer from high false positive rates, wherein benign activities are misclassified as threats or anomalies. This issue arises due to the rigid nature of rule-based detection mechanisms, which may trigger alerts based on benign variations in network traffic patterns. The high volume of false positives can overwhelm security analysts and lead to alert fatigue, reducing the efficacy of threat detection.

3.2 Proposed System

In the proposed system of network traffic analysis, we advocate for the integration of Machine Learning (ML) techniques to overcome the limitations of traditional rule-based and signature-based approaches. Our proposed system aims to leverage ML algorithms to enhance the efficiency, accuracy, and adaptability of network traffic analysis, thereby improving network security and performance management. At the core of our proposed system is the utilization of ML algorithms for traffic classification, intrusion detection, anomaly detection, and predictive analysis. By training ML models on labeled datasets containing diverse network traffic patterns, our system can autonomously learn to distinguish between normal and malicious activities, identify emerging threats, and predict future network behavior. This shift from manual rule-based analysis to automated ML-driven analysis enables real-time threat detection and response, reducing reliance on human intervention and mitigating the risk of alert fatigue. Furthermore, our proposed system incorporates advanced preprocessing techniques for feature extraction and data normalization, facilitating the effective training of ML models on complex network data. By extracting relevant features from raw network traffic data, our system can capture meaningful patterns and relationships that are indicative of security threats or performance issues. This preprocessing step is essential for optimizing model performance and enhancing the accuracy of threat detection and classification.

ADVANTAGES:

The proposed system of network traffic analysis, integrating Machine Learning (ML) techniques, offers several advantages over traditional rule-based and signature-based approaches:

1. **Enhanced Accuracy:** ML algorithms can analyze vast amounts of network data and learn complex patterns, enabling more accurate detection of security threats and performance anomalies. By leveraging advanced algorithms and sophisticated feature extraction techniques, the proposed system can achieve higher detection rates while minimizing false positives.
2. **Real-time Threat Detection:** ML-based systems enable real-time analysis of network traffic, allowing for the immediate detection and response to security threats as they emerge. By continuously monitoring network activity and adapting to evolving threats, the proposed system enhances the organization's ability to proactively defend against cyber-attacks.

3.3 HARDWARE & SOFTWARE REQUIREMENT SPECIFICATION:

Hardware System Configuration

- Processor - I3 processor or Above
- RAM - 4 GB (min)
- Hard Disk - 40 GB

Software Requirements

- Operating system : Windows 8 Ultimate or above.
- Coding Language : Python.
- Databases : Kaggle Network Traffic Dataset, KDD CUP 1999 Dataset.
- Tools : NumPy, Pandas, Matplotlib, Scikit-learn and TensorFlow

3.4 SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- **ECONOMICAL FEASIBILITY**
- **TECHNICAL FEASIBILITY**
- **SOCIAL FEASIBILITY**

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

4. SYSTEM DESIGN

4.1 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS

The Primary goals in the design of the UML are as follows:

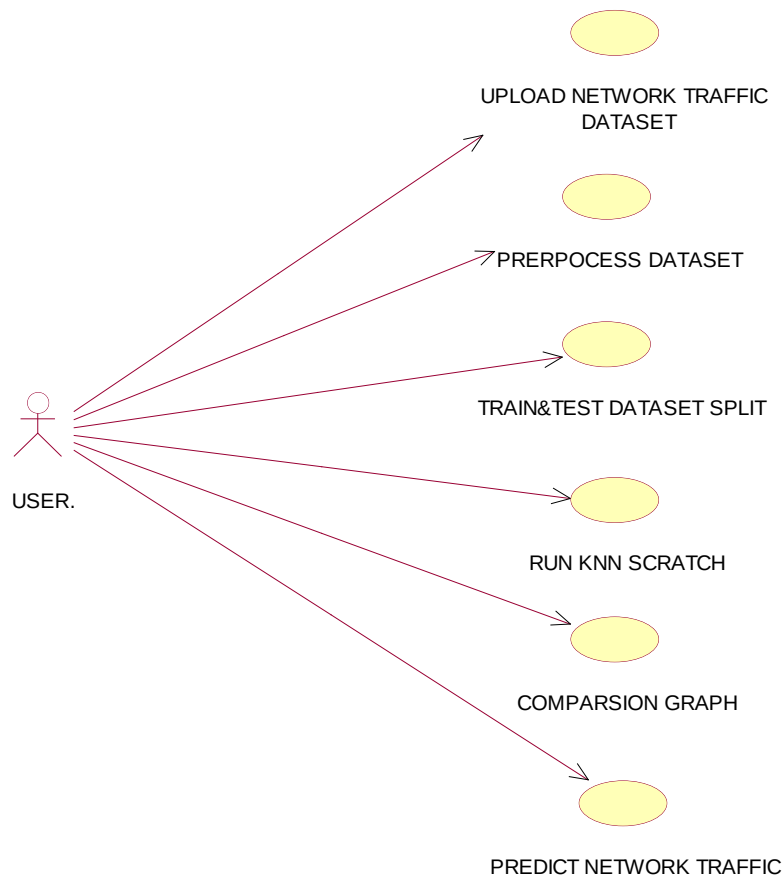
1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.

6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

7. Integrate best practices.

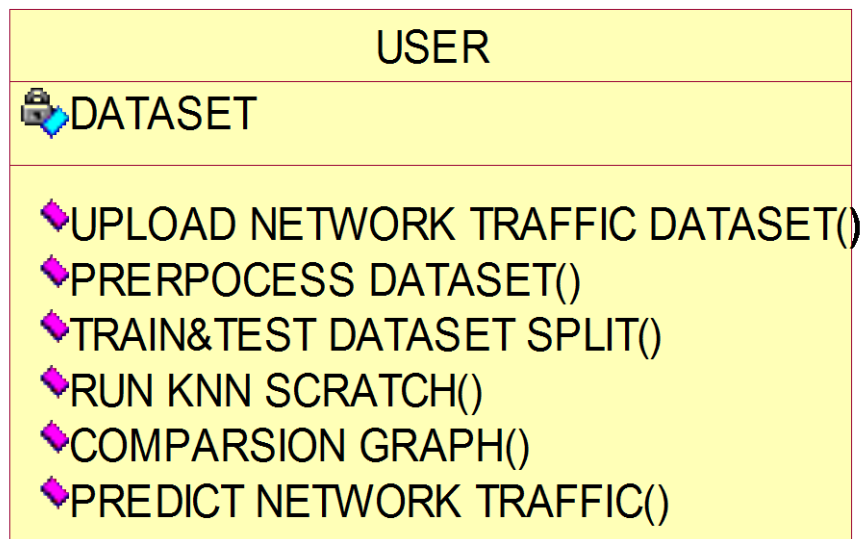
4.2 USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. Fig: USE CASE DIAGRAM



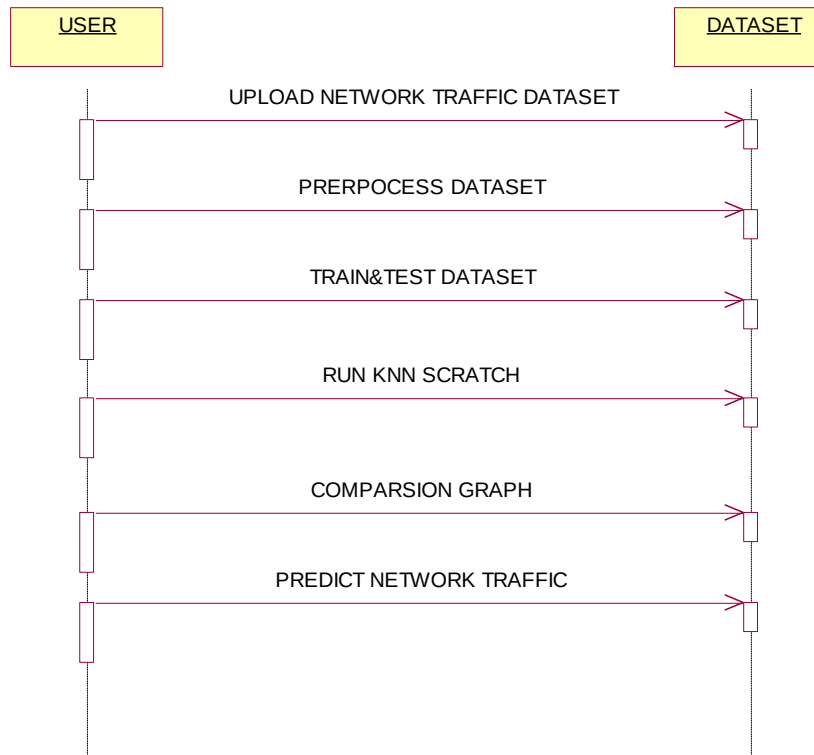
4.3 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information. Fig: CLASS DIAGRAM



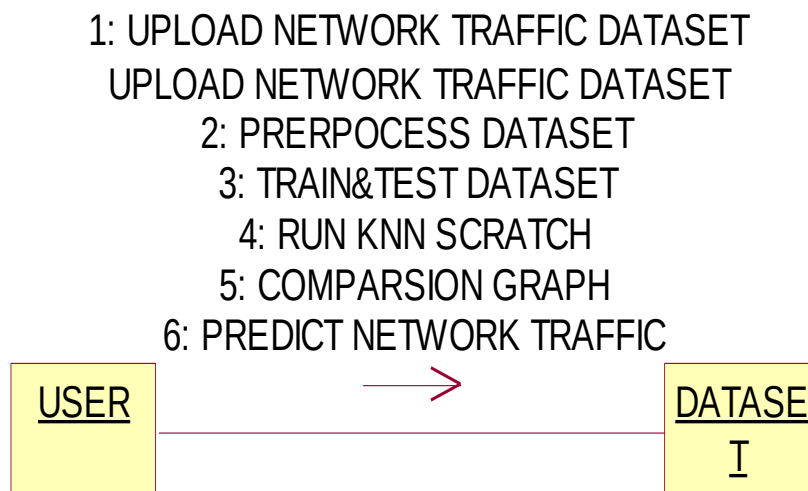
4.4 SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams. Fig: SEQUENCE DIAGRAM



4.5 COLLABRATION DIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. Fig:
COLLABRATION DIAGRAM



5. IMPLEMENTATION

5.1 MODULES:

In this project we are developing carpooling application which can be managed by two users

- 1) Driver: driver can signup and login to application and then enter his location name, latitude and longitude (here note that mobiles will have GPS we can get location from GPS but computer or laptop don't have GPS and we can track from IP but that IP must be registered with DN server and this will not be available so we need to enter location manually) and wait for passenger request. If any passenger in 3 miles distance send request then driver can accept request from multiple users and then click on 'Start Ride' link to start ride and then once ride completed then entered miles travelled to collect fare.
- 2) Passengers: passenger can signup and login to application and then manual enter their location and share with Drivers and drivers can accept request to start ride. Passenger can give ratings also on the scale of 1 to 5

➤ To implement this project we have designed following modules

- 1) Upload Network Traffic Dataset: using this module we will upload dataset to application
- 2) Data Preprocessing: using this module we will clean dataset by removing missing values and ML algorithms will not take characters values so we need to process dataset to convert all non-numeric characters to numeric characters by assigning integer ID to each unique string data. After processing we will split dataset into train and test where 80% dataset will be used for training and 20% for testing
- 3) Run KNN Algorithm: using above train data we will train KNN algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction
- 4) Run Naive Bayes Algorithm: using above train data we will train Naïve Bayes algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction
- 5) Run Decision Tree Algorithm: using above train data we will train Decision tree algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction

- 6) Run SVM Algorithm: using above train data we will train SVM algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction
- 7) Comparison Graph: using this module we will show accuracy graph between all algorithms

SOFTWARE ENVIRONMENT:

5.2 What is Python?

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs are generally smaller than other programming languages like Java.

Programmers have to type relatively little, and the language's indentation requirements keep code readable all the time.

The Python language is used by many tech giants, including Google, Amazon, Facebook, Instagram, Dropbox, and Uber. The biggest strength of Python is a huge collection of standard libraries, which can be used for the following –

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like OpenCV, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library, and it contains code for various purposes like regular

expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

2. Embeddable

Complementary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in another language.

3. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

4. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world

5. Simple and Easy

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy to learn, understand, and code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

6. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. These further aid the readability of the code.

7. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one Free and Open-Source

Like we said earlier, Python is **freely available**. But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

8. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent feature

9. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Any doubt still now about the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python require less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python for beginners.

2. Affordable

Python is free; therefore, individuals, small companies, or big organizations can leverage the freely available resources to build applications. Python is popular and widely used, so it gives you better community support.

The 2019 GitHub annual survey showed us that Python has overtaken Java as the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you

can professionally build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is rarely seen on the **client- side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbon Nelle**.

The reason it is not so famous despite the existence of Bryton is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java Database Connectivity)** and **ODBC (Open Database Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of the Python Programming Language

History of Python

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venner¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum over Wiskunde in Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it." Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So, I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin- end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

Python Development Steps

Guido Van Rossum published the first version of Python code (version 0.9.0) at Altisource's in February 1991. This release included already exception handling, functions, and the core data types of lists, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting Unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x. The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it" Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "//" can be used to have the "old" behavior.
- Text Vs. Data Instead of Unicode Vs. 8-bit

Purpose

We demonstrated that our approach enables successful segmentation of intra retinal layers even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports

multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code.

Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project

TensorFlow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

NumPy

NumPy is a general-purpose array-processing package. It provides a high- performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object

- Sophisticated(broadcasting)functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, NumPy can also be used as an efficient multi- dimensional container of generic data. Arbitrary data-types can be defined using NumPy which allows NumPy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and I Python_shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

Scikit– learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use.

Install Python Step-by-Step in Windows

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So, the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheat sheet here. The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: **<https://www.python.org>**



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.



Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	 Download	Release Notes
Python 3.6.9	July 2, 2019	 Download	Release Notes
Python 3.7.3	March 25, 2019	 Download	Release Notes
Python 3.4.10	March 18, 2019	 Download	Release Notes
Python 3.5.7	March 18, 2019	 Download	Release Notes
Python 2.7.16	March 4, 2019	 Download	Release Notes
Python 3.7.2	Dec. 24, 2018	 Download	Release Notes

Step4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files					
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671a5b2db4aef7b9ab01b0f09be	23017663	SG
XZ compressed source tarball	Source release		d33e4aa66097051c2eca45ee3604803	17131432	SG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583daf1a442cbalce08e6	34898416	SG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a4b773bf5e4a936b241f	28082845	SG
Windows .help file	Windows		d63999573a2c96b2ac56cade6b47cd2	8131761	SG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64/x64	9b00c3cf6d9ec0b9ab6e3184ae0728a2	7504391	SG
Windows x86-64 executable installer	Windows	for AMD64/EM64/x64	a702b4b0ad76db9d83043a583e563400	2688368	SG
Windows x86-64 web-based installer	Windows	for AMD64/EM64/x64	28cb1c50f86d72a8e53a3bd351b4bd2	1362904	SG
Windows x86 embeddable zip file	Windows		9fab3bd18b41879fda94133574139d8	6741828	SG
Windows x86 executable installer	Windows		33cc602942a54446ad86451478394789	25663848	SG
Windows x86 web-based installer	Windows		1b670cfa5cd317d82c30983ea371d87c	1324608	SG

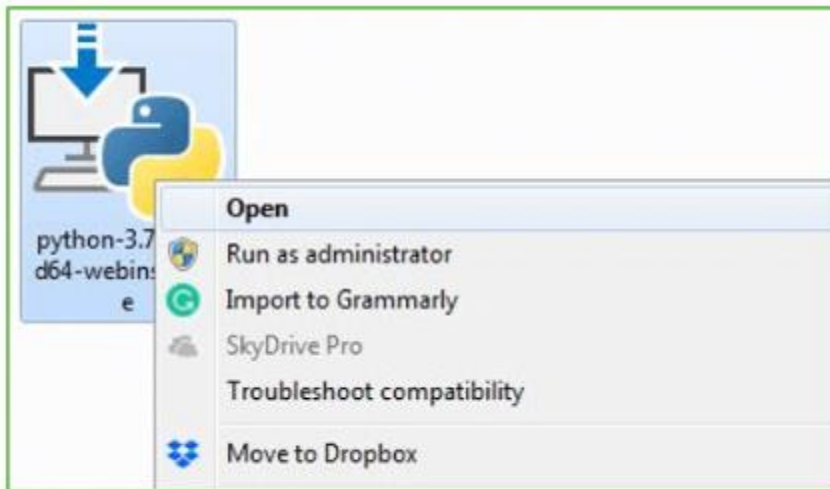
- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, make sure to put a tick on Add Python3.7 to PATH



Step3: Click on Install NOW After the installation is successful. Click on Close.

With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes. Verify the Python Installation



Step1: Click on Start

Step2: In the Windows Run Command, type “cmd”.



Step3: Open the Command prompt option.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\DELL>python -V
Python 3.7.4

C:\Users\DELL>_
```

Step4: Let us test whether the python is correctly installed. Type **python -V** and press Enter.

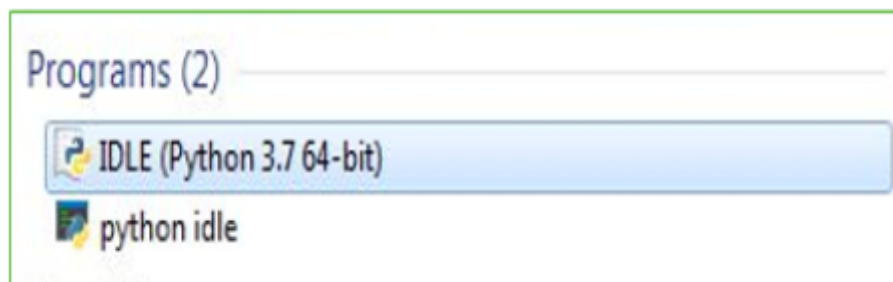
Step5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

Check how the Python IDLE works

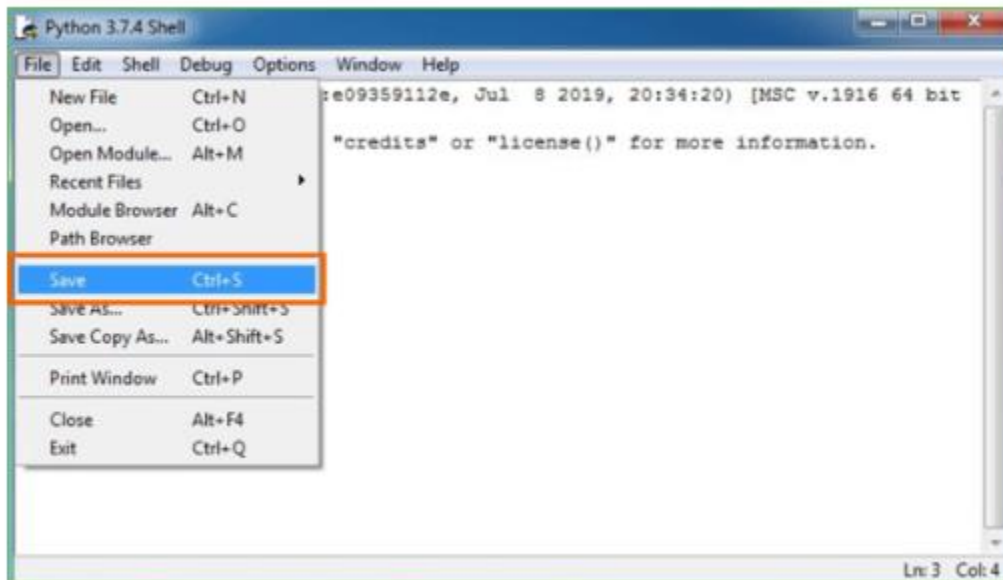
Step1: Click on Start

Step2: In the Windows Run command, type “python idle”.



Step3: Click on IDLE(Python3.764-bit) and launch the program

Step4: To go ahead with working in IDLE you must first save the file. **Click on File**



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. **enter print**

5.3 What is Machine Learning?

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning: -

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction*. Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are –

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data

points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML –

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term “**Machine Learning**” in 1959 and defined it as a “**Field of study that gives computers the capability to learn without being explicitly programmed**”.

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to Indeed, Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of **\$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is [Python](#)! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as [Keras](#), [TensorFlow](#), [Scikit-learn](#), etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as [Fork Python](#) available Free on GeeksforGeeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model** – A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature** – A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label)** – A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training** – The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction** – Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning** – This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.
- **Unsupervised Learning** – This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning** – This involves using unlabelled data like Unsupervised

Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.

- **Reinforcement Learning** – This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As [ML algorithms](#) gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning: -

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it. Coding

5.4 Coding:

File Name: Main.py

```
from tkinter import messagebox
from tkinter import *
from tkinter import simpledialog
import tkinter
from tkinter import filedialog
from tkinter.filedialog import askopenfilename
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.preprocessing import LabelEncoder
```

```

from sklearn.preprocessing import normalize
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
import os
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn import svm
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score

global filename

global X,Y
global dataset
global main
global text
accuracy = []
global X_train, X_test, y_train, y_test

main = tkinter.Tk()
main.title("Network Traffic Analysis Using Machine Learning") #designing main screen
main.geometry("1300x1200")

#traffic names VPN and NON-VPN
class_labels = ['BROWSING', 'CHAT', 'FT', 'MAIL', 'P2P', 'STREAMING', 'VOIP', 'VPN-
BROWSING', 'VPN-CHAT', 'VPN-FT', 'VPN-MAIL', 'VPN-P2P',
                'VPN-STREAMING', 'VPN-VOIP']

#function to upload dataset
def uploadDataset():
    global filename
    global dataset
    text.delete('1.0', END)
    filename = filedialog.askopenfilename(initialdir="Dataset")
    text.insert(END,filename+" loaded\n\n")

```

```

dataset = pd.read_csv(filename)
text.insert(END,"Dataset before preprocessing\n\n")
text.insert(END,str(dataset.head()))
label = dataset.groupby('class1').size()
label.plot(kind="bar")
plt.show()

#function to perform dataset preprocessing
def DataPreprocessing():
    global X,Y
    global dataset
    global X_train, X_test, y_train, y_test
    text.delete('1.0', END)
    #replace missing values with 0
    dataset.fillna(0, inplace = True)
    le = LabelEncoder()
    #traffic type selection
    dataset['class1'] = pd.Series(le.fit_transform(dataset['class1'].astype(str)))
    text.insert(END,"Dataset after preprocessing\n\n")
    text.insert(END,str(dataset.head()))

    temp = dataset.values
    X = temp[:,0:dataset.shape[1]-1] #taking X and Y from dataset for training
    Y = temp[:,dataset.shape[1]-1]
    X = normalize(X)
    indices = np.arange(X.shape[0])
    np.random.shuffle(indices)
    X = X[indices]
    Y = Y[indices]
    text.insert(END,"Dataset after features normalization\n\n")
    text.insert(END,str(X)+"\n\n")
    text.insert(END,"Total records found in dataset : "+str(X.shape[0])+"\n")
    text.insert(END,"Total features found in dataset: "+str(X.shape[1])+"\n\n")
    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)
    text.insert(END,"Dataset Train and Test Split\n\n")
    text.insert(END,"80% dataset records used to train ML algorithms :
"+str(X_train.shape[0])+"\n")
    text.insert(END,"20% dataset records used to train ML algorithms :
"+str(X_test.shape[0])+"\n")

```

```

def runKNN():
    global X,Y
    global X_train, X_test, y_train, y_test
    global accuracy
    accuracy.clear()
    text.delete('1.0', END)
    cls = KNeighborsClassifier(n_neighbors = 2)
    cls.fit(X, Y)
    predict = cls.predict(X_test)
    a = accuracy_score(y_test,predict)*100
    accuracy.append(a)
    precision = precision_score(y_test, predict,average='macro') * 100
    recall = recall_score(y_test, predict,average='macro') * 100
    fmeasure = f1_score(y_test, predict,average='macro') * 100
    text.insert(END,"KNN Accuracy : "+str(a)+"\n")
    text.insert(END,"KNN Precision : "+str(precision)+"\n")
    text.insert(END,"KNN Recall   : "+str(recall)+"\n")
    text.insert(END,"KNN FScore   : "+str(fmeasure)+"\n\n")

```

```

def runNB():
    global X,Y
    global X_train, X_test, y_train, y_test
    global accuracy
    cls = GaussianNB()
    cls.fit(X_train, y_train)
    predict = cls.predict(X_test)
    a = accuracy_score(y_test,predict)*100
    accuracy.append(a)
    precision = precision_score(y_test, predict,average='macro') * 100
    recall = recall_score(y_test, predict,average='macro') * 100
    fmeasure = f1_score(y_test, predict,average='macro') * 100
    text.insert(END,"Naive Bayes Accuracy : "+str(a)+"\n")
    text.insert(END,"Naive Bayes Precision : "+str(precision)+"\n")
    text.insert(END,"Naive Bayes Recall   : "+str(recall)+"\n")
    text.insert(END,"Naive Bayes FScore   : "+str(fmeasure)+"\n\n")

```

```

def runDT():
    global X_train, X_test, y_train, y_test

```

```

global accuracy
cls = DecisionTreeClassifier()
cls.fit(X_train, y_train)
predict = cls.predict(X_test)
a = accuracy_score(y_test,predict)*100
accuracy.append(a)
precision = precision_score(y_test, predict,average='macro') * 100
recall = recall_score(y_test, predict,average='macro') * 100
fmeasure = f1_score(y_test, predict,average='macro') * 100
text.insert(END,"Decision Tree Accuracy : "+str(a)+"\n")
text.insert(END,"Decision Tree Precision : "+str(precision)+"\n")
text.insert(END,"Decision Tree Recall   : "+str(recall)+"\n")
text.insert(END,"Decision Tree FScore   : "+str(fmeasure)+"\n\n")

```

```
def runSVM():
```

```

    global X_train, X_test, y_train, y_test
    global accuracy
    cls = svm.SVC()
    cls.fit(X_train, y_train)
    predict = cls.predict(X_test)
    a = accuracy_score(y_test,predict)*100
    accuracy.append(a)
    precision = precision_score(y_test, predict,average='macro') * 100
    recall = recall_score(y_test, predict,average='macro') * 100
    fmeasure = f1_score(y_test, predict,average='macro') * 100
    text.insert(END,"SVM Accuracy : "+str(a)+"\n")
    text.insert(END,"SVM Precision : "+str(precision)+"\n")
    text.insert(END,"SVM Recall   : "+str(recall)+"\n")
    text.insert(END,"SVM FScore   : "+str(fmeasure)+"\n\n")

```

```
def graph():
```

```

    height = accuracy
    bars = ('KNN Accuracy','Naive Bayes Accuracy','Decision Tree Accuracy','SVM Accuracy')
    y_pos = np.arange(len(bars))
    plt.bar(y_pos, height)
    plt.xticks(y_pos, bars)
    plt.title("Accuracy Comparison Graph")

```

```

plt.show()

def GUI():
    global main
    global text
    font = ('times', 16, 'bold')
    title = Label(main, text='Network Traffic Analysis Using Machine Learning')
    title.config(bg='darkviolet', fg='gold')
    title.config(font=font)
    title.config(height=3, width=120)
    title.place(x=0,y=5)

    font1 = ('times', 12, 'bold')
    text=Text(main,height=30,width=110)
    scroll=Scrollbar(text)
    text.configure(yscrollcommand=scroll.set)
    text.place(x=10,y=100)
    text.config(font=font1)

    font1 = ('times', 13, 'bold')
    uploadButton = Button(main, text="Upload Network Traffic Dataset",
command=uploadDataset, bg='#ffb3fe')
    uploadButton.place(x=900,y=100)
    uploadButton.config(font=font1)

    processButton = Button(main, text="Data Preprocessing", command=DataPreprocessing,
bg='#ffb3fe')
    processButton.place(x=900,y=150)
    processButton.config(font=font1)

    knnButton = Button(main, text="Run KNN Algorithm", command=runKNN, bg='#ffb3fe')
    knnButton.place(x=900,y=200)
    knnButton.config(font=font1)

    nbButton = Button(main, text="Run Naive Bayes Algorithm", command=runNB, bg='#ffb3fe')
    nbButton.place(x=900,y=250)
    nbButton.config(font=font1)

    dtButton = Button(main, text="Run Decision Tree Algorithm", command=runDT, bg='#ffb3fe')

```

```
dtButton.place(x=900,y=300)
dtButton.config(font=font1)

svmButton = Button(main, text="Run SVM Algorithm", command=runSVM, bg='#ffb3fe')
svmButton.place(x=900,y=350)
svmButton.config(font=font1)

graphButton = Button(main, text="Comparison Graph", command=graph, bg='#ffb3fe')
graphButton.place(x=900,y=400)
graphButton.config(font=font1)

main.config(bg='forestgreen')
main.mainloop()

if __name__ == "__main__":
    GUI()
```

6 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS:

6.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.3 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.
Functions : identified functions must be exercised.
Output : identified classes of application outputs must be exercised.
Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

7. SCREENSHOTS

Network Traffic Analysis using ML

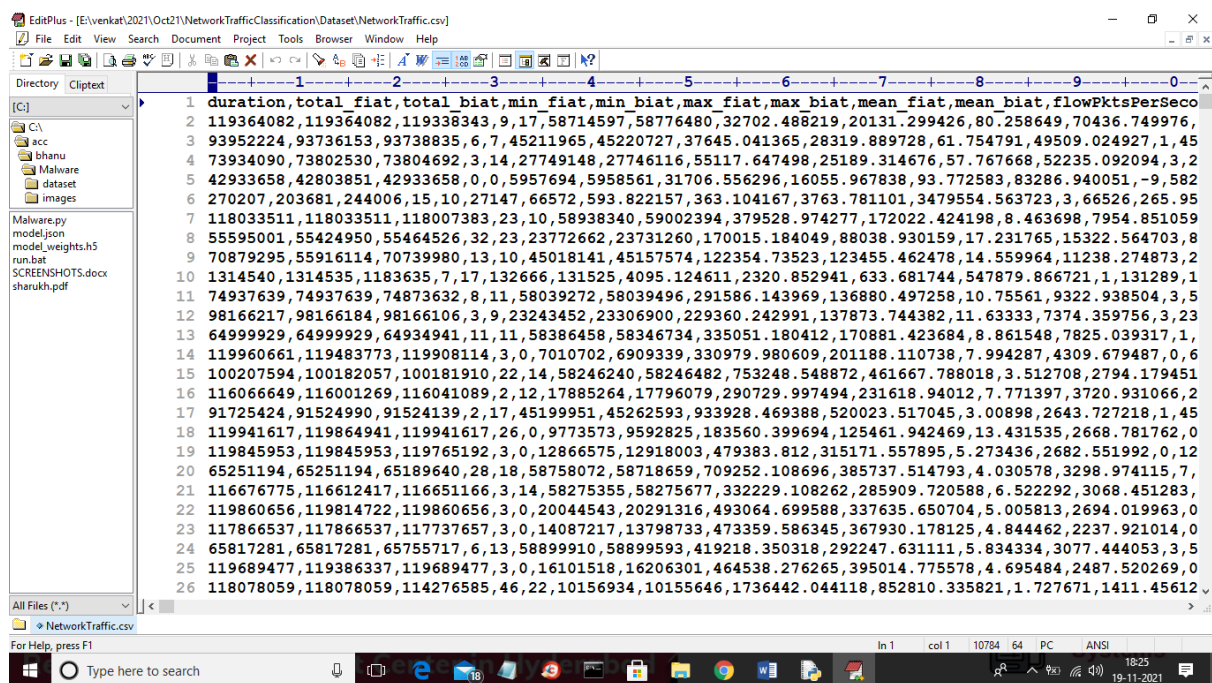
In this project we are using various machine learning algorithms such as KNN, SVM, Decision Tree and Naïve Bayes to predict traffic or classify type of network data such as BROWSING traffic, MAIL traffic etc. Lots of network traffic type of data is available but in this project we are training ML algorithms to predict or classify 14 different types of traffic. Below are the names of such traffic types:-

['BROWSING', 'CHAT', 'FT', 'MAIL', 'P2P', 'STREAMING', 'VOIP', 'VPN-BROWSING', 'VPN-CHAT', 'VPN-FT', 'VPN-MAIL', 'VPN-P2P', 'VPN-STREAMING', 'VPN-VOIP']

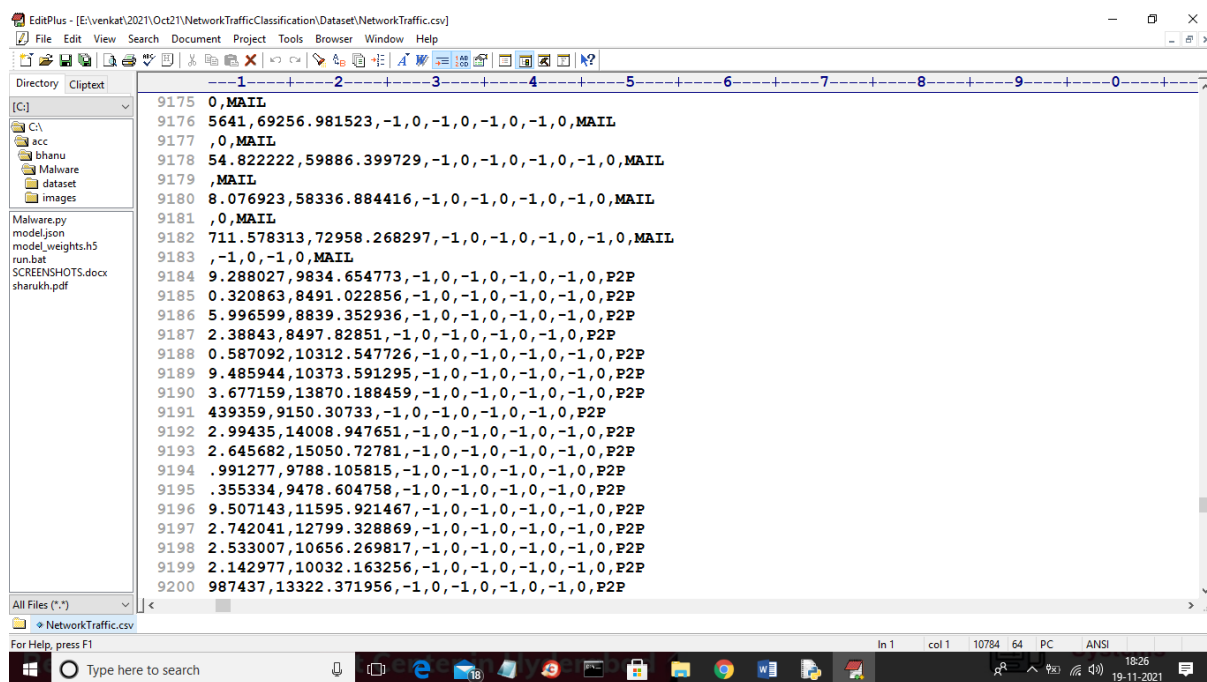
We have evaluated performance of each algorithm in terms of accuracy, precision, recall and FSCORE and in all algorithms KNN shows better result. We downloaded dataset from below link

<https://www.unb.ca/cic/datasets/vpn.html>

below is the dataset screen shot+++++



In above dataset first row represents dataset column names and remaining rows represents dataset values and in last column we can see type of traffic which you can see in below screen.



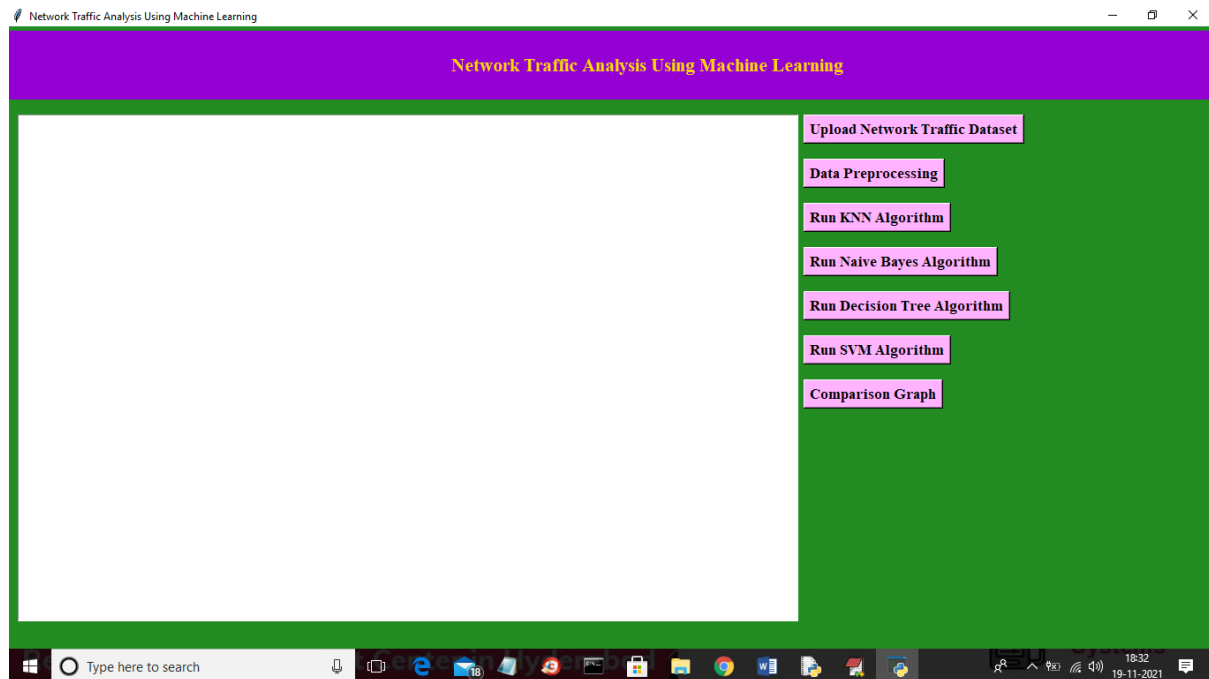
In last column you can see each record is associated with one label such as traffic type.

To implement this project we have designed following modules

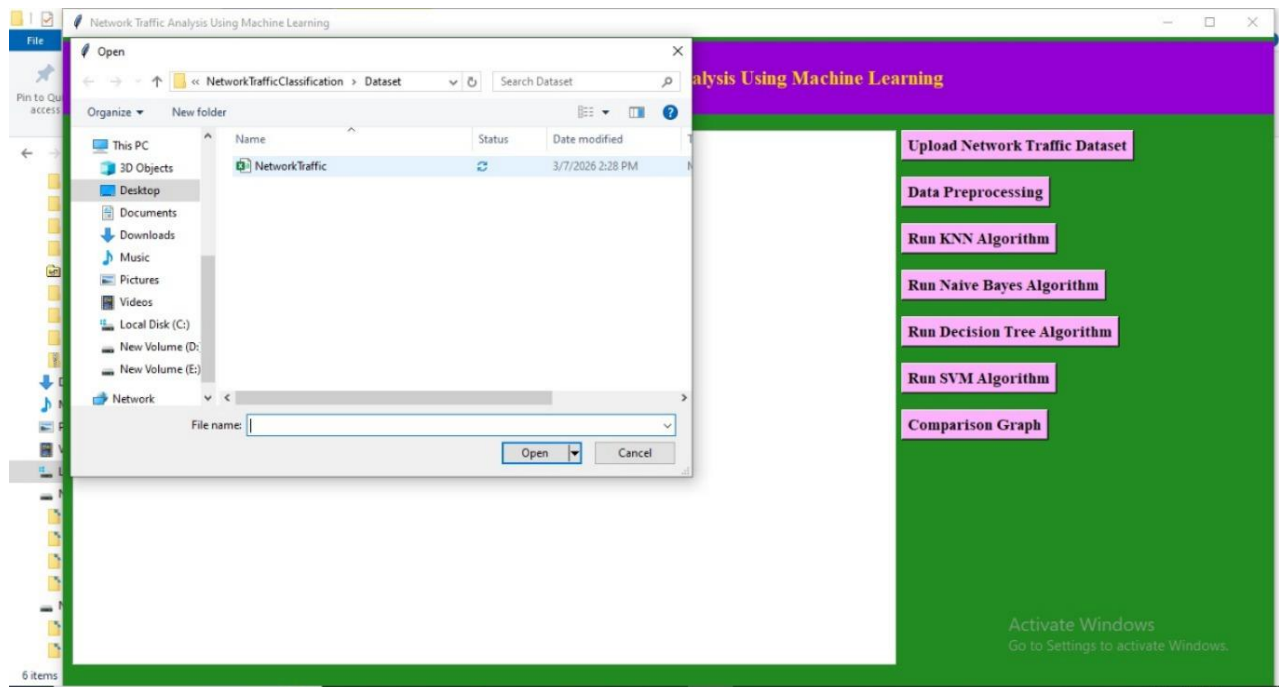
- 1) Upload Network Traffic Dataset: using this module we will upload dataset to application.
- 2) Data Preprocessing: using this module we will clean dataset by removing missing values and ML algorithms will not take characters values so we need to process dataset to convert all non-numeric characters to numeric characters by assigning integer ID to each unique string data. After processing we will split dataset into train and test where 80% dataset will be used for training and 20% for testing.
- 3) Run KNN Algorithm: using above train data we will train KNN algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction.
- 4) Run Naive Bayes Algorithm: using above train data we will train Naïve Bayes algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction.
- 5) Run Decision Tree Algorithm: using above train data we will train Decision tree algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction.
- 6) Run SVM Algorithm: using above train data we will train SVM algorithm and this trained model will be applied on test data to predict traffic type and then calculate accuracy of correct prediction.
- ~~7) Comparison Graph: using this module we will show accuracy graph between all algorithms.~~

SCREEN SHOTS

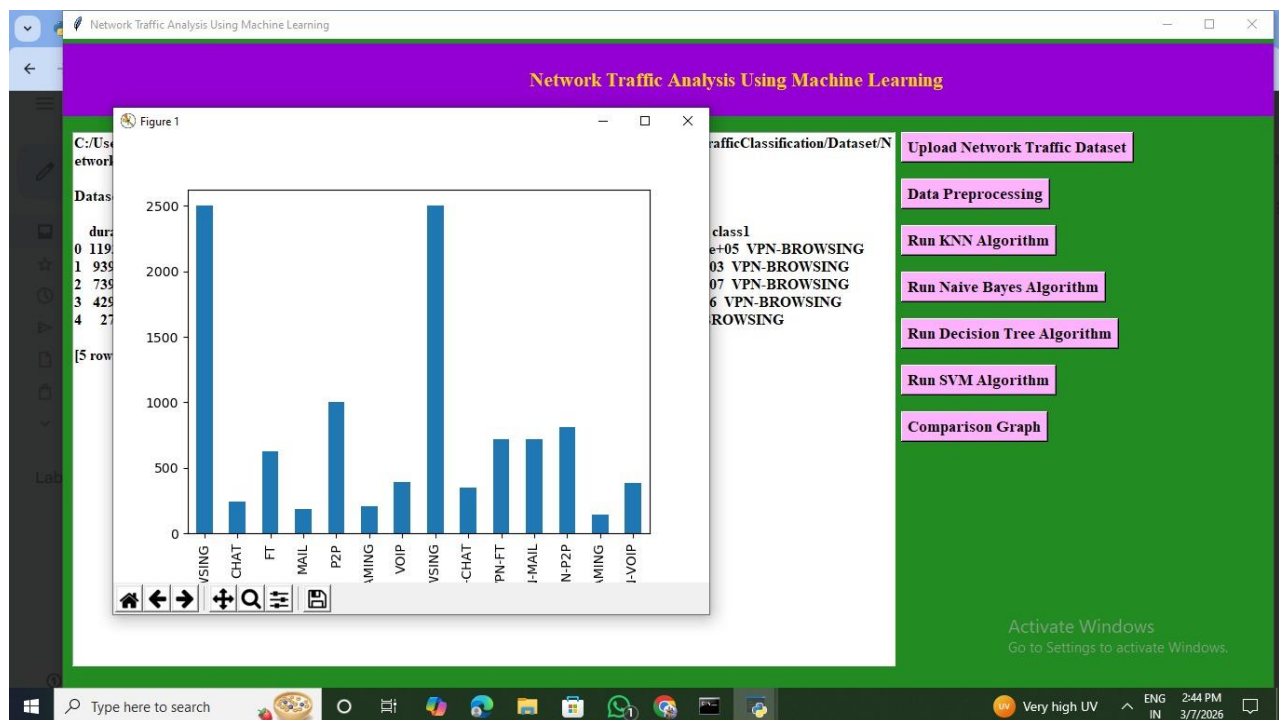
To run project double click on 'run.bat' file to get below screen.



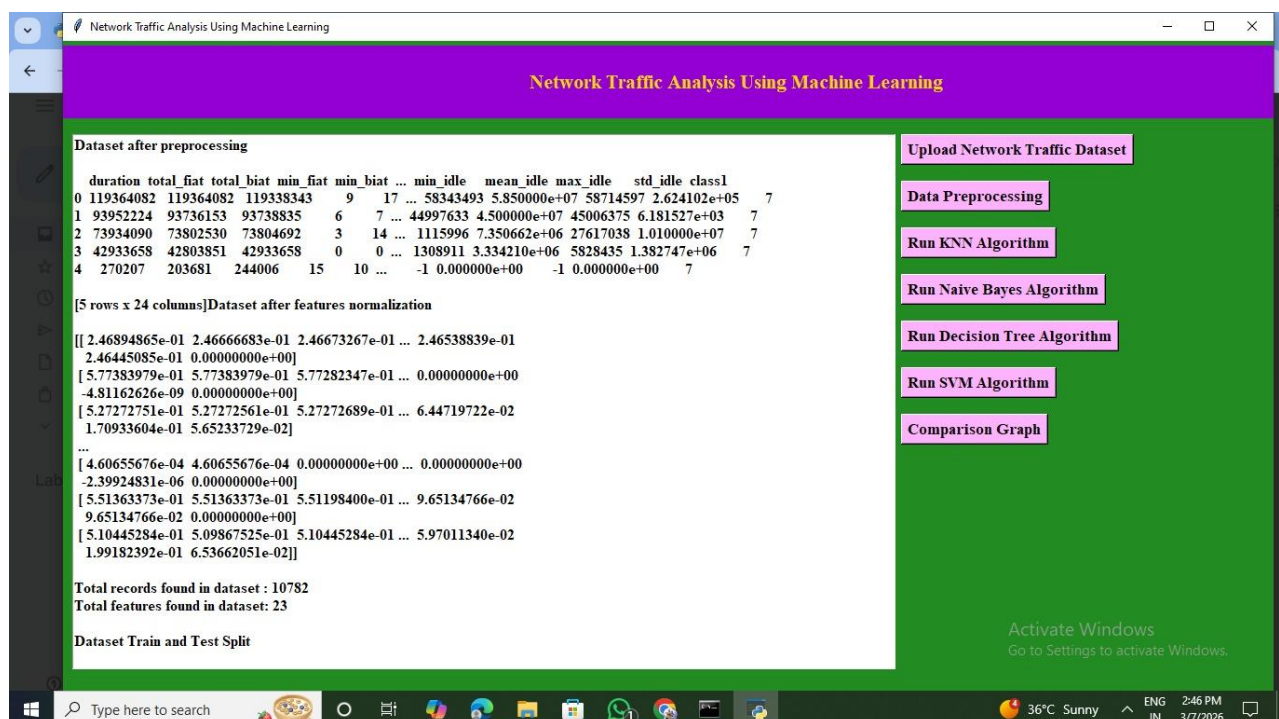
In above screen click on 'upload Network Traffic Dataset' button to upload dataset



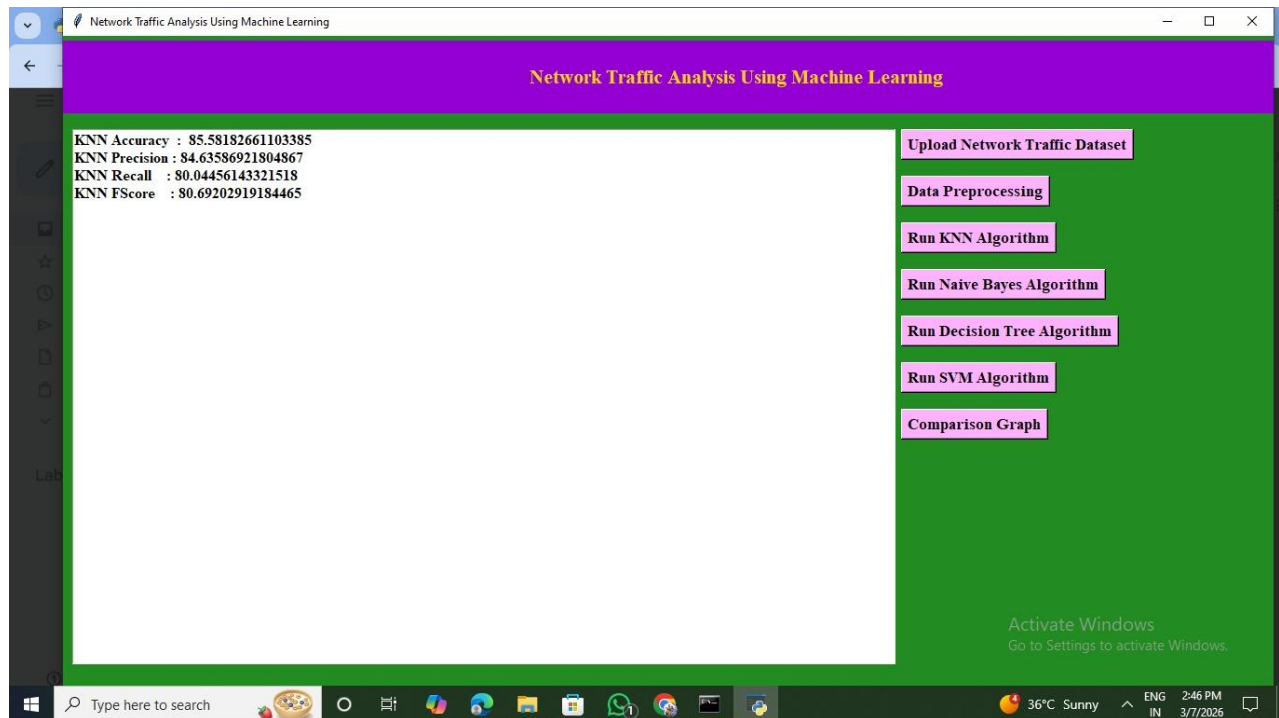
In above screen selecting and uploading 'NetworkTraffic.csv' file and then click on 'Open' button to load dataset and to get below screen.



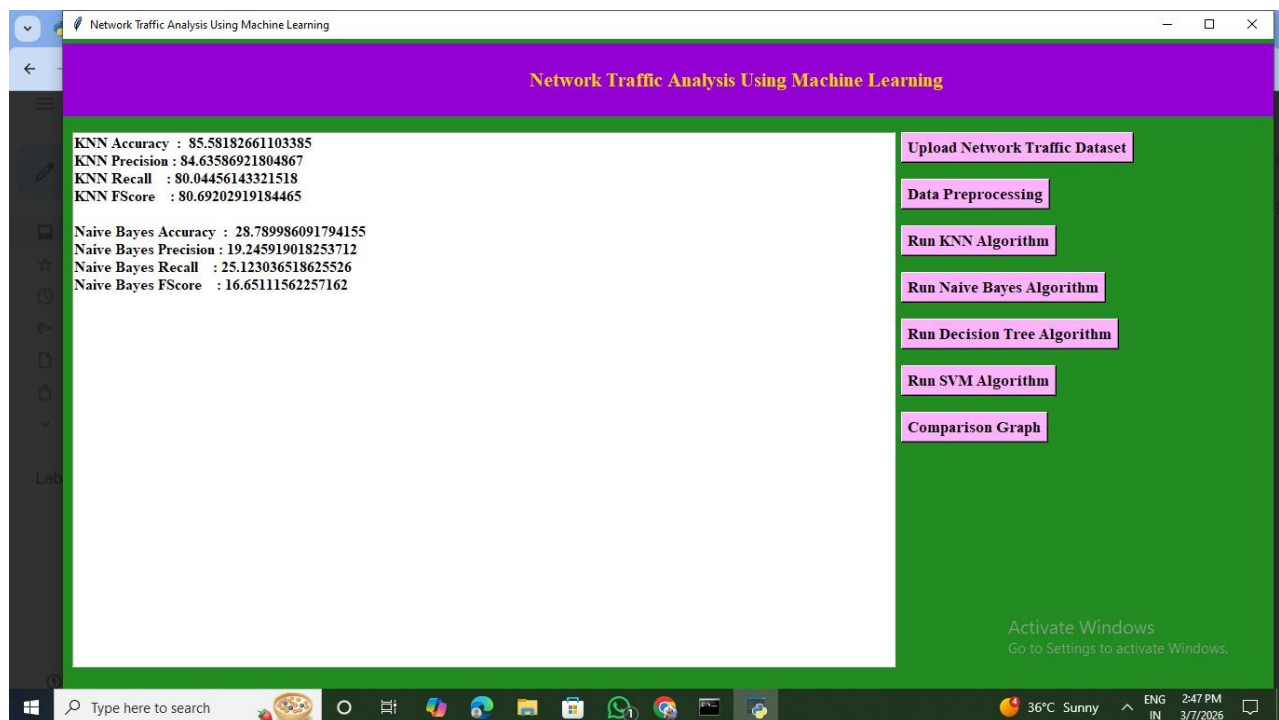
In above screen we can see dataset loaded and dataset contains lots of non-numeric values so we need to process it and in graph x-axis we can see traffic type and y-axis represents total records in dataset for that traffic. Now close above graph and then click on 'Data Preprocessing' to clean dataset.



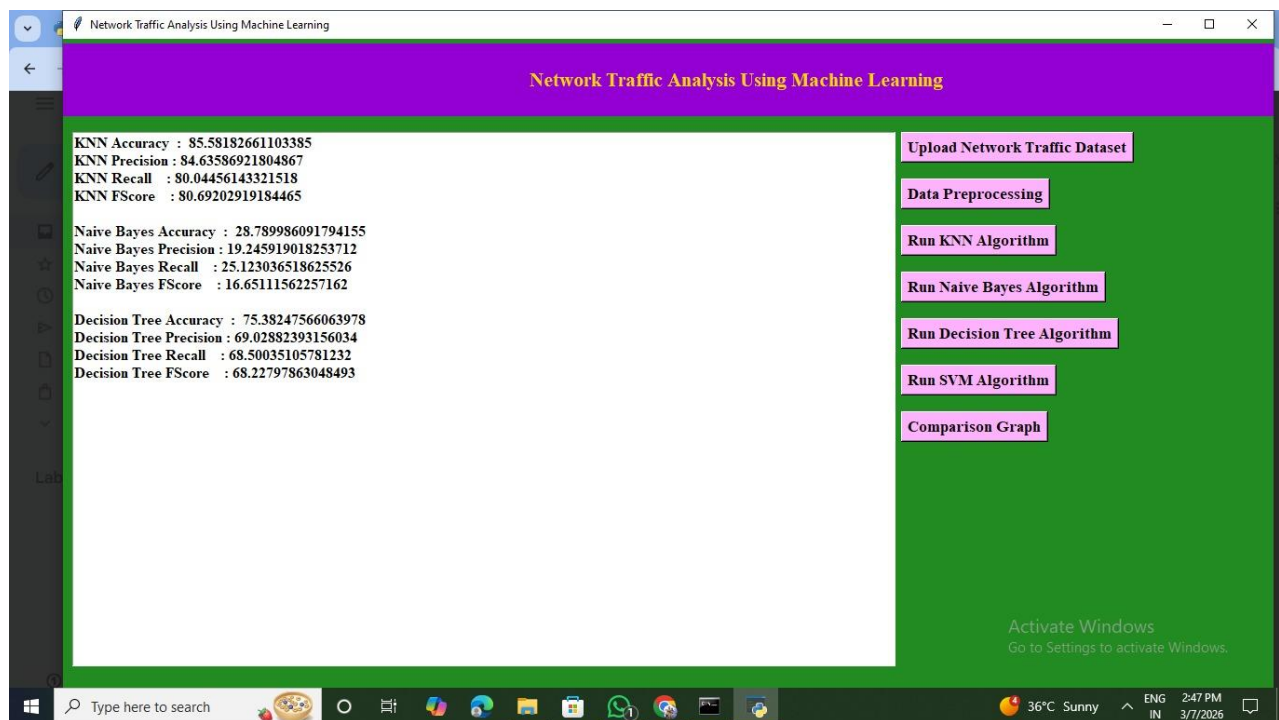
In above screen we can see all dataset converted to numeric format and in last we can see total records and columns found in dataset and then splitted dataset percentage for train and test records. Now train and test data is ready and now click on 'Run KNN Algorithm' button to train KNN and get below result.



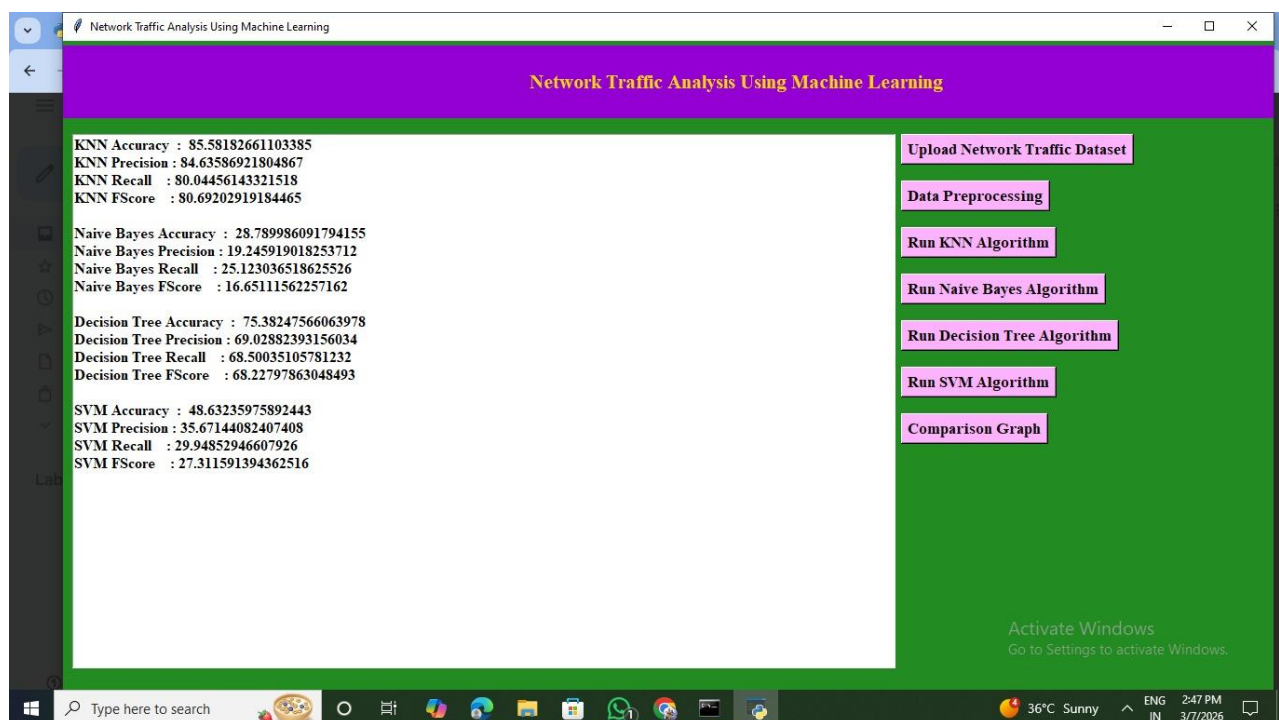
In above screen with KNN we got 84% accuracy and now click on 'Run Naïve Bayes Algorithm' button to train it



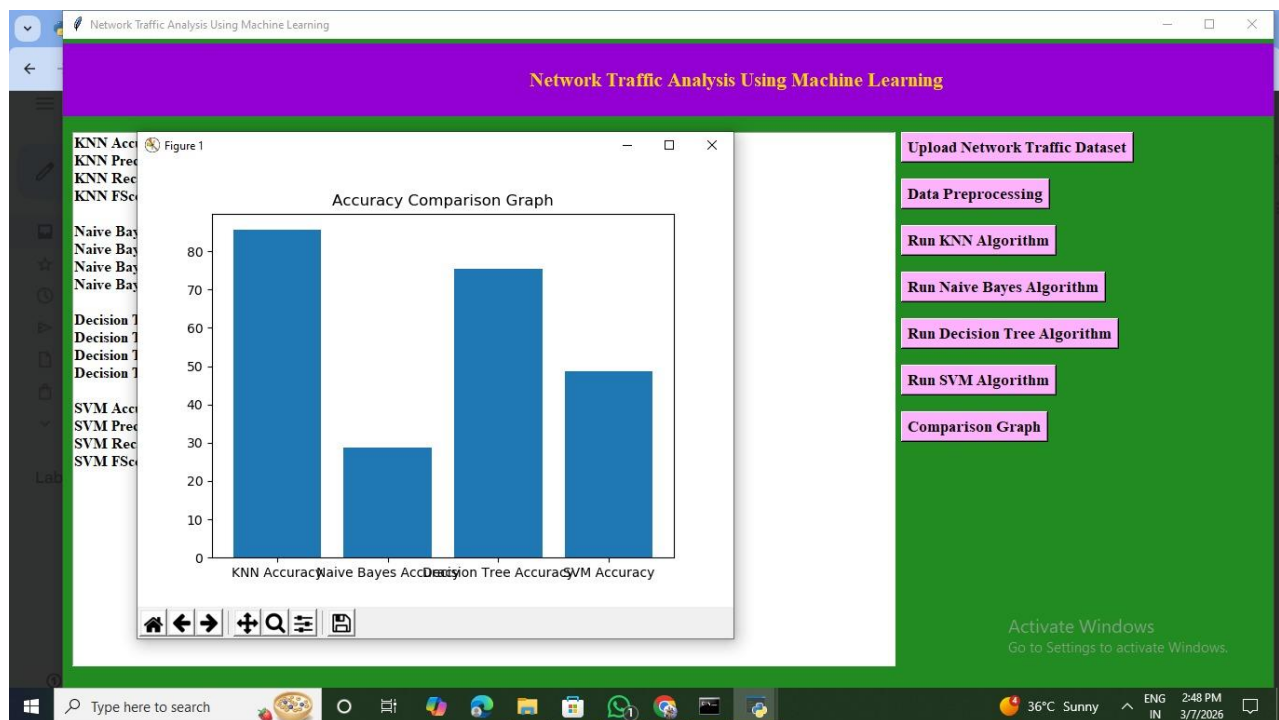
In above screen for same dataset with naïve bayes we got 28% accuracy and now click on 'Run Decision Tree Algorithm' button to get below result.



In above screen for same dataset with SVM we got 74% accuracy and now click on ‘Run SVM Algorithm’ button to get below result.



In above screen with SVM we got 50% accuracy and now click on ‘Comparison Graph’ button to get below result.



In above graph x-axis represents algorithm name and y-axis represents accuracy of those algorithms and in all algorithms KNN shows better result.

8.CONCLUSION

In conclusion, the integration of Machine Learning (ML) techniques into network traffic analysis represents a transformative approach to enhancing network security and performance management in today's interconnected digital landscape. Throughout this paper, we have explored the evolution of network traffic analysis from traditional rule-based methods to advanced ML-driven approaches, highlighting the advantages, challenges, and opportunities associated with this paradigm shift. ML-based network traffic analysis offers several significant benefits, including enhanced accuracy, real-time threat detection, adaptability to emerging threats, and scalability. By leveraging ML algorithms, organizations can achieve higher detection rates, reduce false positives, and automate the process of threat detection and response. Moreover, ML-based systems enable predictive analysis, allowing organizations to anticipate potential security threats and performance issues before they occur. However, the adoption of ML in network traffic analysis also presents challenges, such as the need for labeled datasets, the complexity of feature engineering, and concerns regarding model interpretability and transparency. Addressing these challenges requires interdisciplinary collaboration, innovative research, and the development of robust methodologies and best practices for ML-based network traffic analysis.

9.REFERENCES

Certainly! Here are some references that you can use for your paper on Network Traffic Analysis using ML:

1. Papagiannaki, K., & Iannaccone, G. (2007). Practical Network Traffic Analysis and Intrusion Detection Using Machine Learning. *IEEE Transactions on Signal Processing*, 55(8), 3967-3976.
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7. Le, A., Gascón, A., & Plaza, A. (2019). A Survey of Machine Learning Techniques Applied to Anomaly Detection on Host-Level Data. *IEEE Access*, 7, 181742-181771.
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10. Khan, A. N., & Bashir, F. (2019). A Review of Anomaly Detection Techniques in Network Traffic. *Journal of Cybersecurity*, 5(1), tyz003.

These references cover a range of topics related to network traffic analysis using Machine Learning, including intrusion detection, anomaly detection, and data mining techniques. They should provide a solid foundation for your paper and support your arguments effectively.